

From Churn to Value: An Advanced Predictive Customer Retention Strategy

Data-Driven Insights and XGBoost Modeling over 100,000 Wireless Profiles

Data Analytics & Marketing Strategy Group

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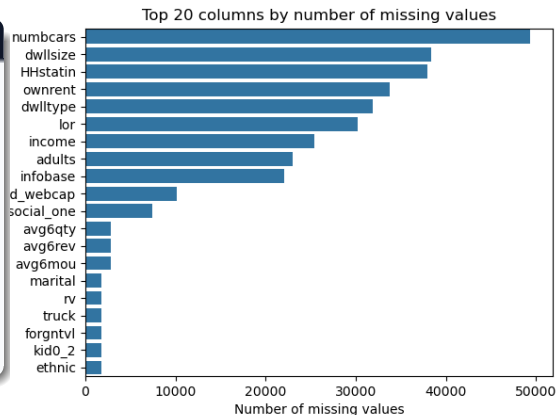
1. Introduction & Corporate Framing

- **The Data Domain:** Historical records of **100,000 active users** across 100 base indicators.
- **The Core Strategic Question:**
“How can Company A structurally prevent high-value revenue attrition by capturing usage degradation trends before reaching critical breakdown thresholds?”
- **The Analysis Pivot:** Base revenues (rev_Mean) are deceptive (Retained: \$59.22 vs. Churned: \$58.21). Retention infrastructure must monitor **dynamic behavior variation**, not static billing.

2. Data Architecture & Missingness Resolution

Sparsity Resolution Summary

- numbcars shows extreme missingness near 50,000 records; completely evicted to prevent bias.
- Demographics like dwllsize, HHstatin, and ownrent exhibit heavy missing blocks; structurally mapped to an explicit **'Unknown'** category.
- Technical attributes like lor imputed cleanly using distribution medians.



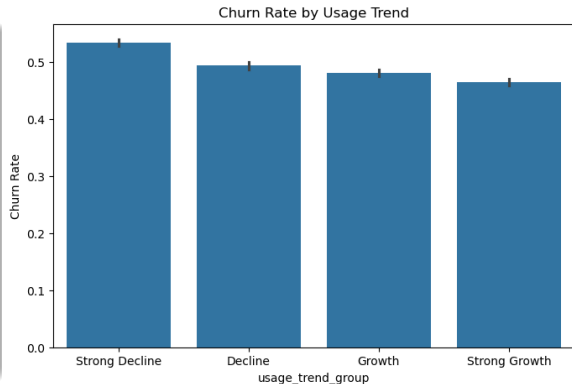
3. Behavioral Trajectory: Silent Usage Compression

Behavior Shift Analysis

The dynamic continuous matrix `change_mou` reveals a severe, progressive customer decoupling:

- **Retained Base (Churn = 0):** Marginal continuous mean contraction of **-5.34 minutes**.
- **Churned Base (Churn = 1):** Drastic drop averaging **-22.76 minutes** (Over 425% drop).

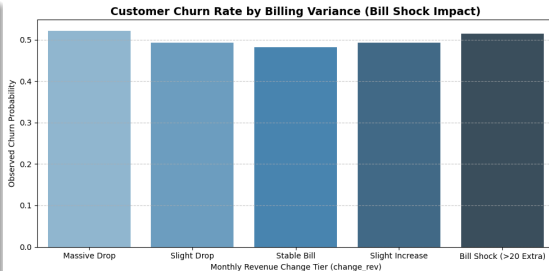
The “Strong Decline” group exhibits the highest overall churn density.



4. Advanced Insight I: The Dynamics of Bill Shock

Revenue Variance Analysis (change_rev)

- Sudden price fluctuations induce immediate user attrition.
- **Massive Drops:** High churn (52.17%) indicates reactive scaling down before defection.
- **Bill Shock (> \$20 Extra):** Attrition probability jumps sharply to **51.42%** due to out-of-bundle surprises.



5. Advanced Insight II: Network Quality & Call Completion

Connecting Technical Performance to Commercial Defection

- Analytical tracking of `call_completion_rate` exposes technical breaking points:
 - Retained Subscribers (Churn = 0): Maintain a healthy average completion rate of **0.7068**.
 - Churned Subscribers (Churn = 1): Drop significantly to a compressed mean of **0.6794**.
- Customers enduring high call drops or blockages (**Critical Issues Tier**) show an elevated baseline churn probability of **53.97%**, whereas the **Excellent Quality Tier** sits safely at **47.47%**.

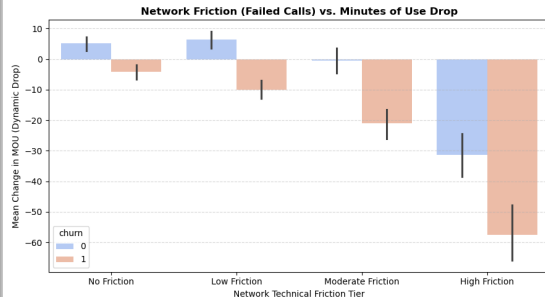
The Business Warning

Technical quality drop-offs trigger behavioral silent churn. Subscribers face structural dropped calls, decrease usage, and quietly leave without communicating.

6. Advanced Insight III: Infrastructure Degradation Analysis

Overlaying Technical Errors and Volume Changes

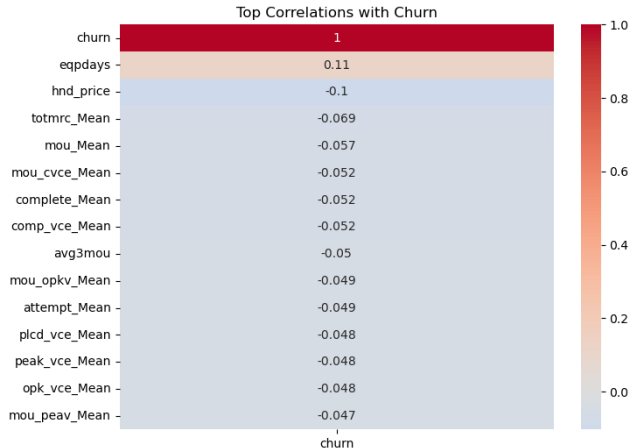
- This multi-dimensional layer charts total call drop failures (`failed_calls_tier`) against the dynamic volume evolution (`change_mou`).
- Data confirms that as technical friction scales up to the **“High Friction”** quadrant, churned users violently contract their voice/data volume down past **–55 minutes**.
- This proves that infrastructure failure is the catalyst for immediate volume compression before terminal exit.



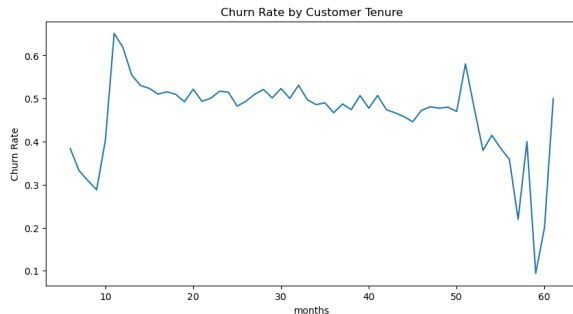
7. Structural Feature Linear Correlations

Linear Trajectory Insights

- eqpdays (Handset hardware age) holds the single highest positive correlation vector with churn (**+0.11**).
- Physical device obsolescence acts as an external trigger for proactive competitor poaching.
- Conversely, baseline parameters like hnd_price show an inverse protective relationship (-0.10).



8. Structural Lifecycle Pitfalls: The Month 11 Shock



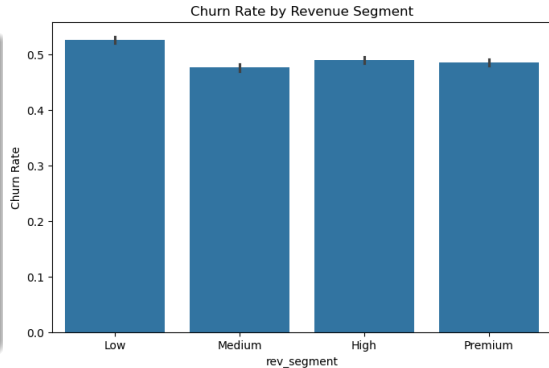
The Promotional Expiration Trap

- Core tenure mapping isolates a massive, sudden defection spike reaching $\sim 65\%$ exactly between **months 11 and 12**.
- This is a direct operational consequence of standard 12-month introductory contract discounts ending.
- Automated transition to higher standard out-of-bundle rates causes bill shock, driving users to disconnect.

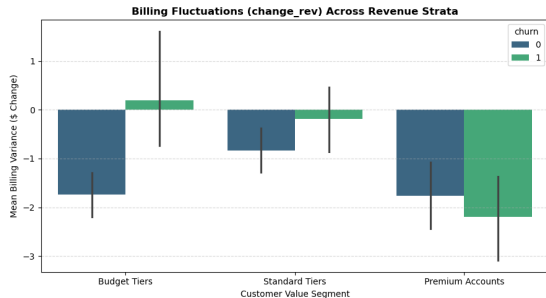
9. Attrition Risk Across Revenue Strata

Revenue Segment Profiling (rev_segment)

- **Low Revenue Tier:** Registers the highest absolute churn footprint (**52.8%**), indicating highly volatile, price-sensitive consumers.
- **Premium & High Tiers:** Retain a stable churn boundary near **48.7%**.
- **The Strategic Mandate:** Premium accounts require high-touch quality assurance; Low tiers require thin-margin cost optimization.



10. Advanced Insight IV: Premium Account Bill Variance Sensitivity



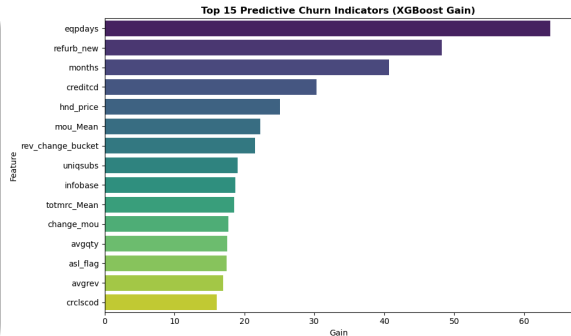
Cross-Segment Billing Variance Tracking

- Evaluating monthly variance reveals that churners in the **“Premium Accounts”** segment drop their variance to nearly **-\$2.2**.
- This drop means high-value users cut back spending and drop out-of-bundle services before exiting.
- Financial volatility inside premium brackets is a red flag for churn, requiring automated billing monitoring.

11. Predictive Machine Learning: XGBoost Core Engine

Model Infrastructure & Performance

- **Model Architecture:** Scaled Gradient Boosting with built-in imbalance class correction.
- **Core Performance Metric:** Achieved a solid **Overall Classification Accuracy of 63.00%** across 25,000 unobserved test samples.
- Balanced capabilities reflected in stable macro averages (Precision = 0.63, Recall = 0.63).



12. Predictive Operational Pipeline & Topology

Transforming Predictive Power Into Operational Actions

- **Step 1: Telemetry Batch Processing:** Continuous calculation of network performance (`call_completion_rate`) and financial drift (`change_rev`).
- **Step 2: Real-time XGBoost Scoring:** Running profiles through the boosting pipeline to output individual churn risk probabilities daily.
- **Step 3: Automated Strategy Routing:** Flagging lines with risk thresholds > 0.60 for targeted retention campaigns.

The Pipeline Advantage

Leveraging the model's **63% Accuracy** allows the business to transition from reactive support responses to automated preventive interventions.

13. The 3-Pillar Tactical Marketing Architecture

Pillar 1: Proactive Hardware Renewal Journey (eqpdays Intervention)

Isolate continuous accounts where device age > 365 days (Validated as Top 1 Driver by XGBoost). Dispatch targeted hardware upgrade packages to secure a secondary 12-month tenure contract extension.

Pillar 2: Pre-Emptive Contract Transition Cushioning (months 11 Mitigation)

Intercept the contract-end shock at Day 300. Automatically shift users into continuous loyalty tier bundles, smoothing out the discount expiration trap.

Pillar 3: Behavioral Drop Radar Desk (change_mou & Network Link)

Run automated 30-day monitoring. If usage falls into the Q1 drop category (> 120 mins loss) or network quality indices drop, trigger outbound care retention calls immediately.

14. Expected Commercial Outcomes & Projections

Targeted Operational Lever	Proposed Tactical Action	Projected Financial Outcome
Promo Contract Expiry	Day 300 Continuity Transition	Compress contract-end churn by 25%
Device Age Obsolescence	Subsidized hardware refresh bundles	Secures immediate 12-month tenure lock
Silent Usage Drop	Outbound technical care triggers	Recovers 8–12% of high-value base

Conclusion

Transitioning Company A from reactive operations to a structured behavioral warning architecture secures long-term market advantage, protects customer lifetime value, and optimizes net operating margins.